



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KSG AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

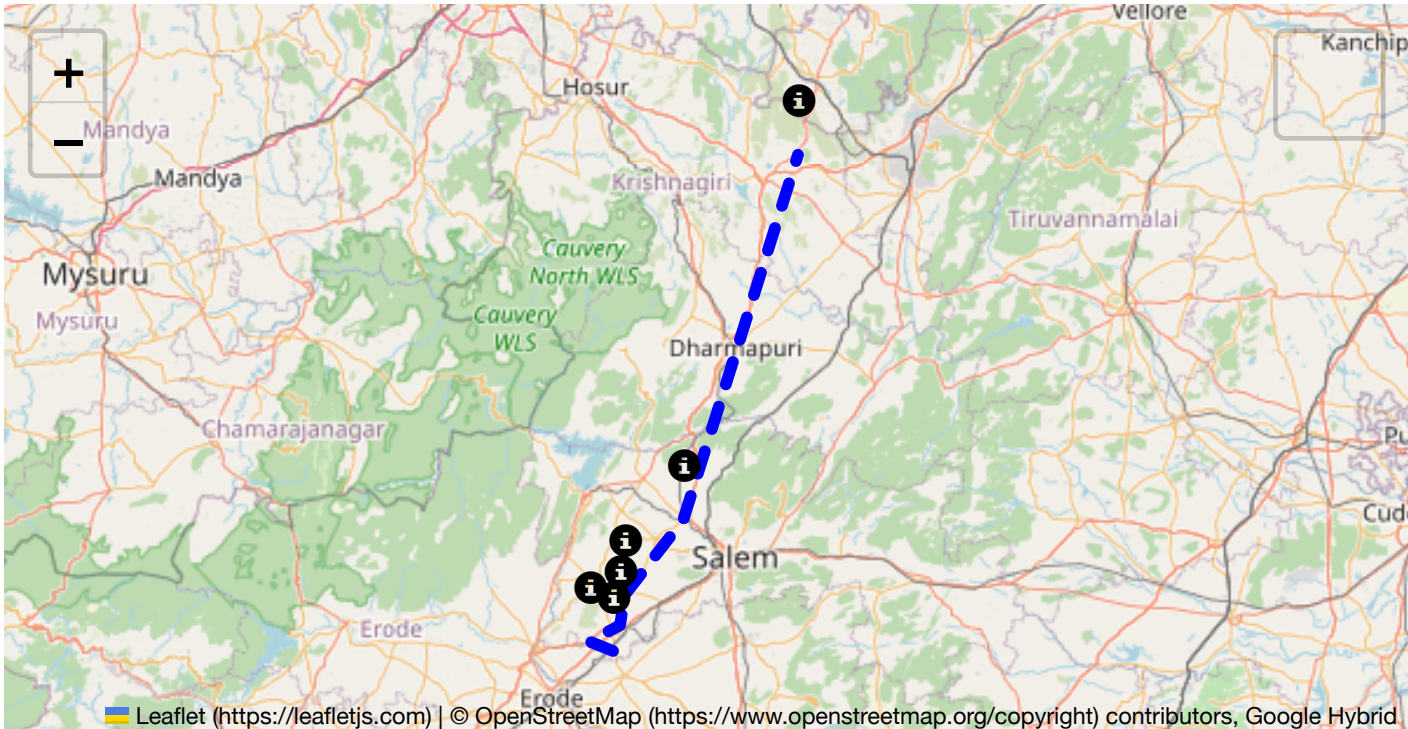
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

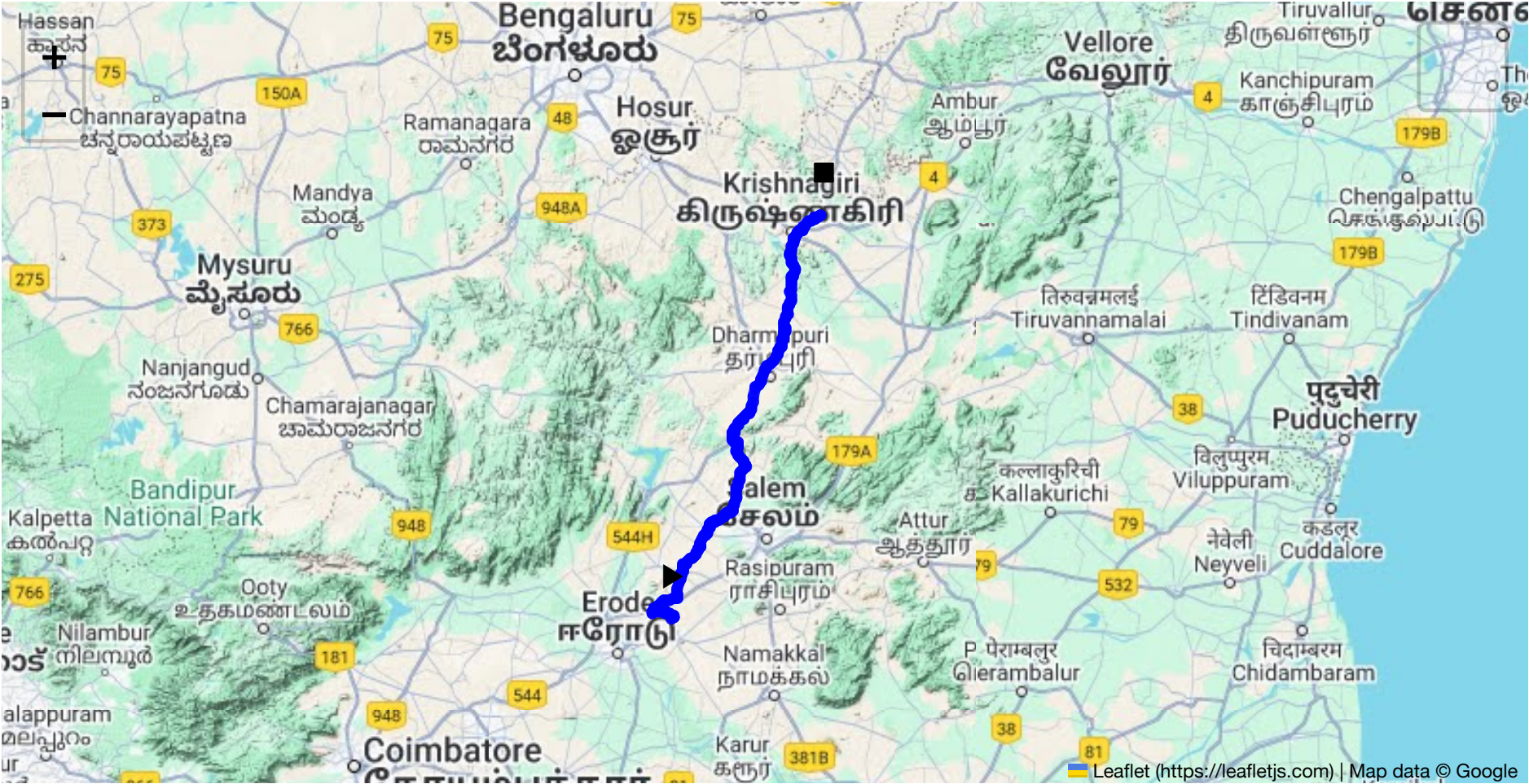
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 161.93 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours

Start: (11.4381, 77.8734)
End: (12.577179, 78.302322)



Welcome to the Journey Risk Management Study

Route Overview: The route covers approximately 161.93 kilometers, beginning at Sangagiri, proceeding through notable waypoints such as Bhavani Main Road, SH 86 in Ayveli, Erumaipatti, and Salem, and concluding at Rayappanur in Krishnagiri. It traverses a mix of state highways and local roads, which are crucial for regional connectivity, particularly for heavy vehicles transporting goods.

Typical Weather Conditions: The weather in Tamil Nadu can vary with the seasons. The area experiences a tropical climate with hot summers (March to May), a wet monsoon season (June to September), and mild winters (November to February). During the monsoon, heavy rainfall can cause flooding, particularly in low-lying areas, and can make driving conditions hazardous due to reduced visibility and slippery roads.

Traffic Patterns: Traffic tends to be heavier during peak hours of 8:00-10:00 AM and 5:00-7:00 PM on weekdays. Salem, being a significant city on this route, often experiences congestion, especially around major intersections and commercial zones. These congestion-prone areas may cause delays, especially during the evening hours.

Road Quality and Infrastructure: The major highways, such as SH 86, are generally in good condition with proper signage and road markings. However, certain stretches may have worn surfaces due to ongoing wear and tear or construction activities. Rural roads connecting smaller towns may have variable conditions and should be navigated with caution, particularly in rainy conditions.

Alternative Routes for Emergencies: In case of road closures or incidents, an alternative route could involve bypassing Salem by taking inner district roads that loop around the city, although these might not support heavy vehicle traffic as effectively. Mobile GPS and navigation apps should be updated to provide real-time alternatives during emergencies.

Local Regulations on Hazardous Materials: Transport of hazardous materials is subject to strict regulations, including obtaining permits from the Tamil Nadu Pollution Control Board and adhering to safety protocols such as using designated routes and stopping only at approved locations.

Historical Incidents: There have been instances of vehicle accidents involving heavy trucks in densely populated areas like Salem, often attributed to human error or mechanical failure. These incidents underscore the importance of adhering to speed regulations and maintaining vehicles properly.

Environmental Considerations: The route passes near agricultural zones and some forested areas. Care must be taken to minimize spillage or leakage of hazardous materials to protect these ecosystems. Awareness and quick response plans are critical in sensitive zones.

Communication Coverage: The route generally has good mobile coverage, though there may be signal weak zones in sparsely populated areas between major towns. Use of satellite communication devices is recommended during travel in these areas to ensure continuous connectivity.

Emergency Response Times: Emergency response services in urban centers like Salem are well-established, with response times typically within 15-30 minutes. In rural areas, the response might be delayed due to distance and road conditions, potentially taking up to an hour or more.

Overall Summary of Risk Assessment: This route is moderately safe for transporting hazardous materials with a well-maintained primary road network and established infrastructure in urban areas. However, careful navigation is necessary during adverse weather, in areas of heavy traffic congestion, and in rural sections with variable road quality. Adhering to local transport regulations and ensuring clear communication channels and emergency plans are recommended to mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.56072, 77.89919	30 KM/Hr
11	Turn	Medium	11.56140, 77.89991	30 KM/Hr
12	Turn	Medium	11.59085, 77.90592	30 KM/Hr
13	Turn	Medium	11.59253, 77.90690	30 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	High	11.64197, 77.95197	15 KM/Hr
16	Turn	Medium	11.64558, 77.95108	30 KM/Hr
17	Turn	Medium	11.67883, 77.96865	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.71972, 78.01202	30 KM/Hr
22	Turn	Medium	11.72765, 78.02291	30 KM/Hr
23	Turn	High	11.73785, 78.03496	15 KM/Hr
24	Turn	High	11.73653, 78.03886	15 KM/Hr
25	Turn	Medium	11.74252, 78.04705	30 KM/Hr
26	Turn	Medium	11.77998, 78.05786	30 KM/Hr
27	Turn	Medium	11.86915, 78.08382	30 KM/Hr
28	Turn	Medium	11.87998, 78.08170	30 KM/Hr
29	Turn	Medium	11.91266, 78.05852	30 KM/Hr
30	Turn	Medium	11.92653, 78.06517	30 KM/Hr
31	Turn	Medium	11.93196, 78.06473	30 KM/Hr
32	Turn	Medium	11.93473, 78.06268	30 KM/Hr
33	Turn	Medium	11.93746, 78.05331	30 KM/Hr
34	Turn	Medium	11.95531, 78.04698	30 KM/Hr
35	Turn	High	11.95854, 78.04990	15 KM/Hr
36	Turn	High	11.96015, 78.04945	15 KM/Hr
37	Turn	Medium	11.96266, 78.05115	30 KM/Hr
38	Turn	High	11.96433, 78.05113	15 KM/Hr
39	Turn	High	11.96542, 78.04938	15 KM/Hr
40	Turn	High	11.96650, 78.04927	15 KM/Hr
41	Turn	Medium	12.17041, 78.16823	30 KM/Hr
42	Turn	Medium	12.22515, 78.19158	30 KM/Hr
43	Turn	Medium	12.22995, 78.19466	30 KM/Hr
44	Turn	Medium	12.24547, 78.19501	30 KM/Hr
45	Turn	Medium	12.24685, 78.19561	30 KM/Hr
46	Turn	Medium	12.25120, 78.20010	30 KM/Hr
47	Turn	Medium	12.25442, 78.20015	30 KM/Hr
48	Turn	Medium	12.29194, 78.19961	30 KM/Hr
49	Turn	Medium	12.32012, 78.20734	30 KM/Hr
50	Turn	Medium	12.40661, 78.22318	30 KM/Hr
51	Turn	Medium	12.42548, 78.22666	30 KM/Hr
52	Turn	Medium	12.45066, 78.21821	30 KM/Hr
53	Turn	Medium	12.50265, 78.22345	30 KM/Hr
54	Turn	High	12.50736, 78.22244	15 KM/Hr
55	Turn	Medium	12.52332, 78.23159	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
56	Turn	High	12.53439, 78.24841	15 KM/Hr
57	Turn	Medium	12.55996, 78.26849	30 KM/Hr
58	Turn	Medium	12.56656, 78.28341	30 KM/Hr
59	Turn	Medium	12.56892, 78.28474	30 KM/Hr
60	Turn	Medium	12.56948, 78.28970	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
31	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
32	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
34	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
35	college	Government Polytechnic College	12.5352064, 78.2460463	30 km/h	Medium
36	college	Government Boys Arts and Science college	12.5323457, 78.2457274	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56140, 77.89991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71972, 78.01202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



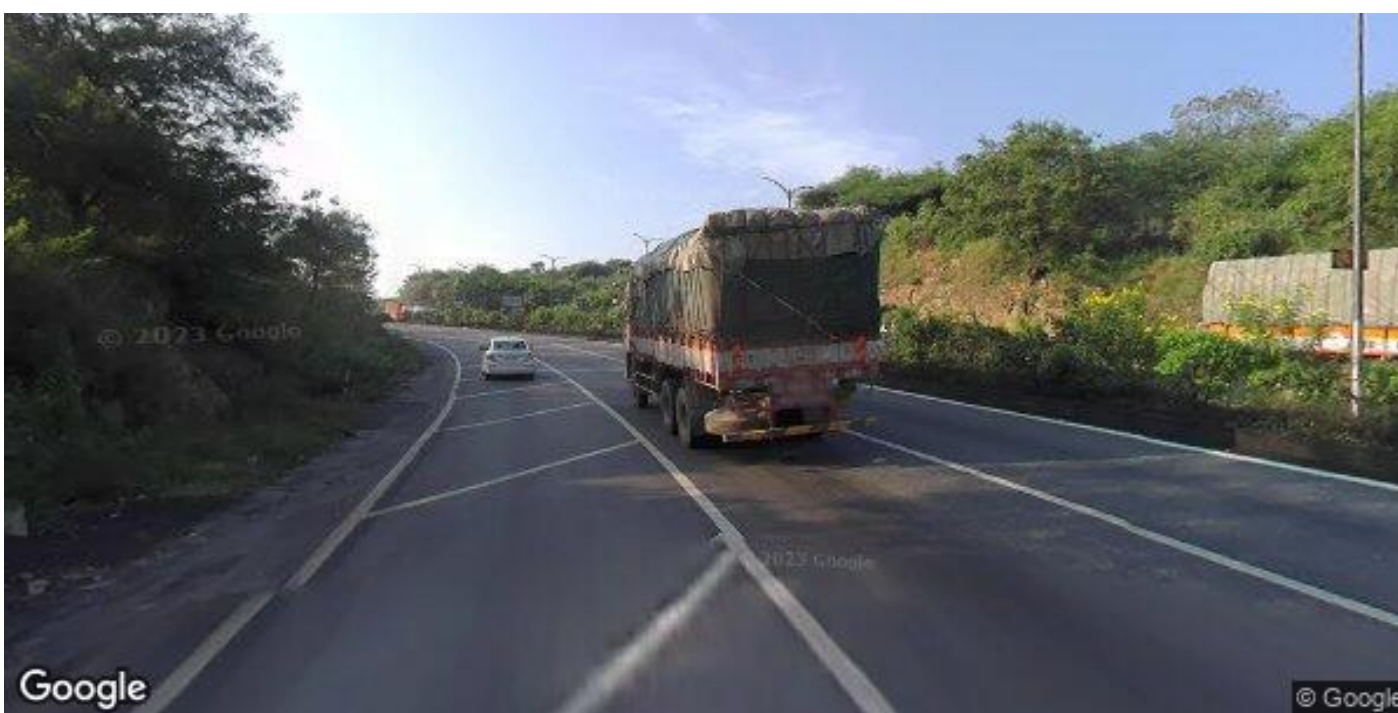
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22515, 78.19158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22995, 78.19466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24547, 78.19501



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24685, 78.19561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25120, 78.20010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25442, 78.20015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29194, 78.19961



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32012, 78.20734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40661, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42548, 78.22666



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45066, 78.21821



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50265, 78.22345



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.50736, 78.22244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52332, 78.23159



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.53439, 78.24841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55996, 78.26849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56656, 78.28341



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56892, 78.28474



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56948, 78.28970

